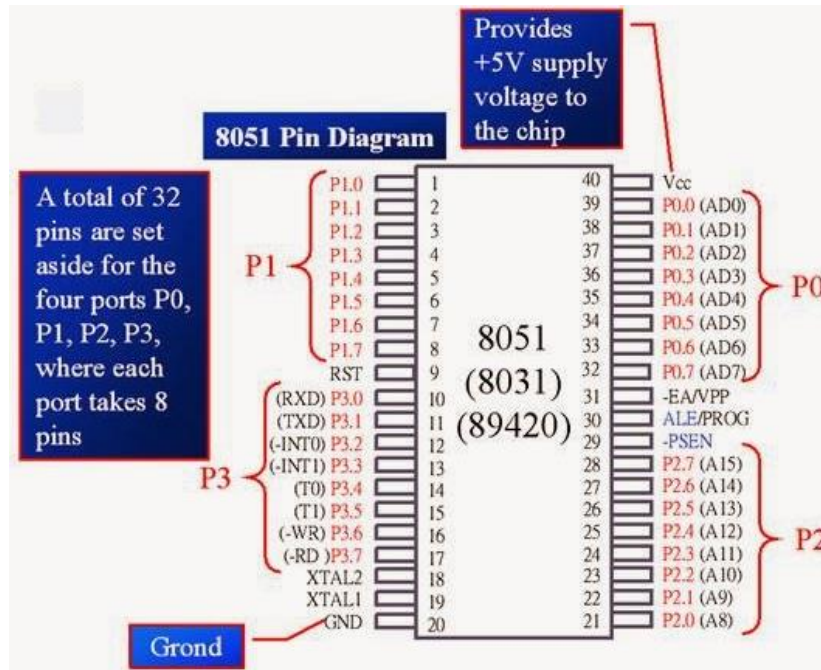


Embedded Computing Principles

Name: Abhay Kumar

Admission No: 25SECE1030034

8051 Microcontroller –Summary



The 8051 is an 8-bit microcontroller developed by Intel. It is one of the most popular microcontrollers used for learning and designing embedded systems. It combines a processor, memory, input/output ports, timers, and communication facilities on a single chip.

The classic 8051 has 4 KB of internal ROM and 128 bytes of internal RAM. ROM is used to store the program instructions, while RAM is used to temporarily store data during program execution. It can also interface with external memory when additional storage is required.

The 8051 has four 8-bit input/output ports named Port 0, Port 1, Port 2, and Port 3. Together, these ports provide 32 programmable I/O pins. These pins can be connected to switches, LEDs, sensors, displays, and other external devices.

The microcontroller contains two 16-bit timers/counters, called Timer 0 and Timer 1. They are used for generating accurate time delays, measuring time intervals, and counting external events. Timers are important in many control and communication applications.

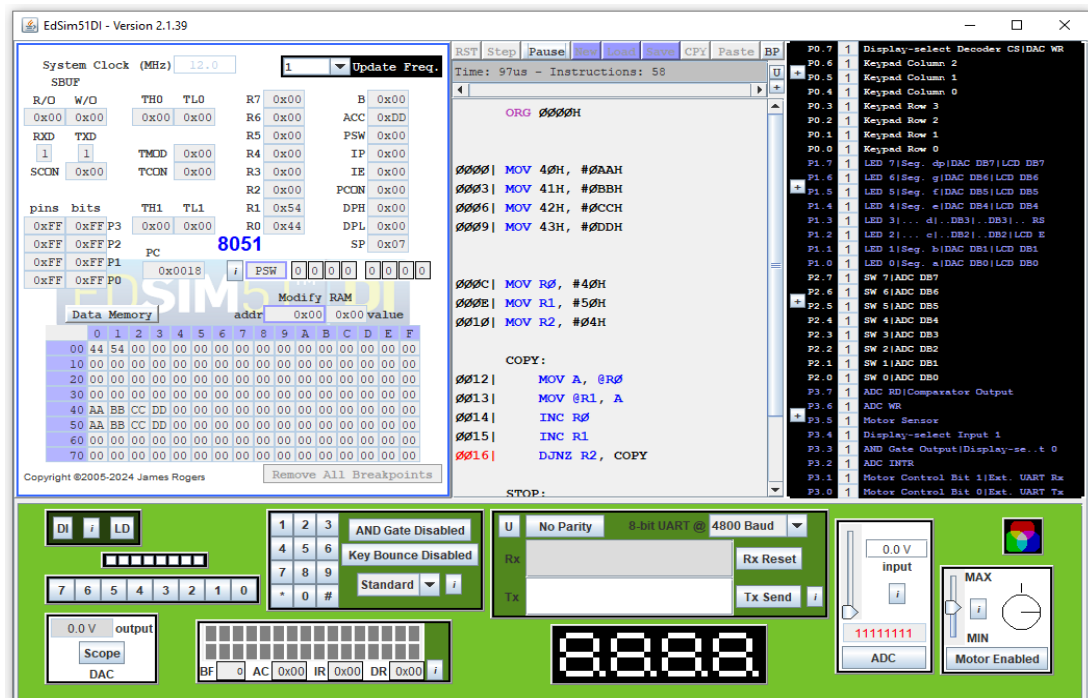
The 8051 also provides serial communication through its built-in serial port. This allows it to exchange data with other microcontrollers, computers, and peripheral devices. It supports interrupt-based operation, which allows the processor to respond quickly to important events.

Important registers in the 8051 include the Accumulator, B register, Program Counter, Stack Pointer, Data Pointer, and Program Status Word. These registers help in performing arithmetic and logical operations, storing temporary data, controlling program execution, and accessing memory.

The 8051 is used in many applications such as industrial automation, traffic-light controllers, motor control systems, keyboards, displays, electronic instruments, and simple robotics. Its ability to control external devices makes it useful in a wide range of embedded applications.

In conclusion, the 8051 is a simple, reliable, and versatile 8-bit microcontroller. Its CPU, memory, I/O ports, timers, serial communication, and interrupt features make it suitable for many control applications. It is also widely used as an introductory microcontroller for understanding embedded-system concepts.

Q. write a program to copy 4 data element from 1 memory block (40H to 45H) to another memory block.



CODE:

```
ORG 0000H
```

```
MOV 40H, #0AAH
```

```
MOV 41H, #0BBH
```

```
MOV 42H, #0CCH
```

```
MOV 43H, #0DDH
```

```
MOV R0, #40H
```

```
MOV R1, #50H
```

```
MOV R2, #04H
```

```
COPY:
```

```
MOV A, @R0
```

```
MOV @R1, A
```

```
INC R0
```

```
INC R1
```

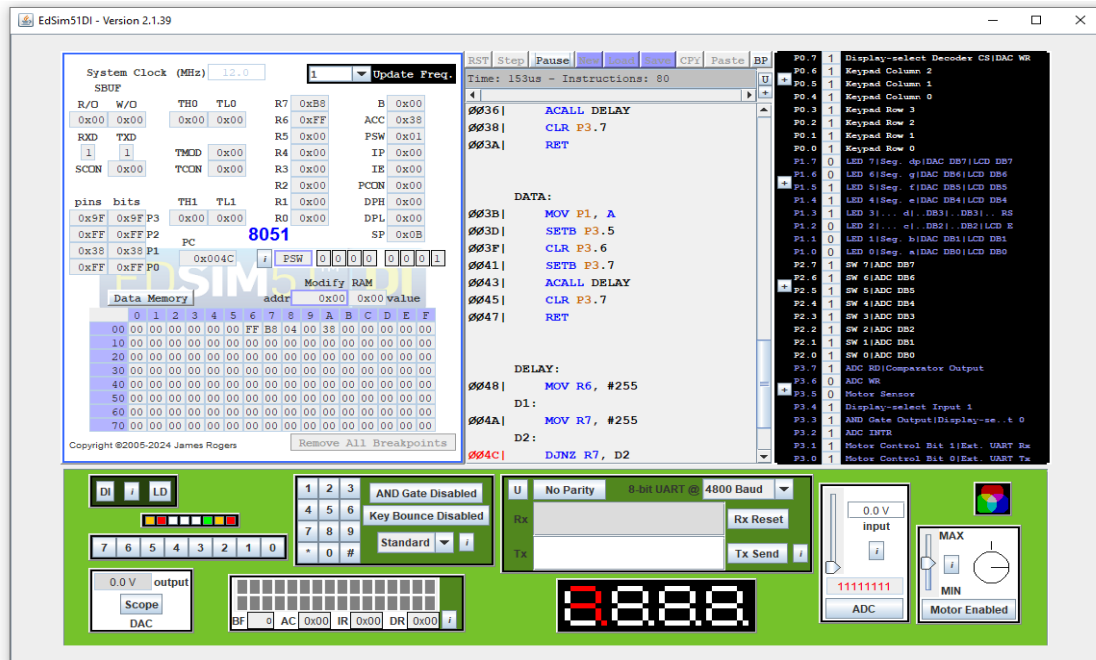
```
DJNZ R2, COPY
```

```
STOP:
```

SJMP STOP

END

Q. write a program to blink the lcd display (2 digit of lcd display)



CODE :

ORG 0000H

MOV A, #38H

ACALL COMMAND

MOV A, #0CH

ACALL COMMAND

BLINK:

MOV A, #80H

ACALL COMMAND

MOV A, #'2'

ACALL DATA

MOV A, #81H

ACALL COMMAND

MOV A, #'5'

ACALL DATA

ACALL DELAY

MOV A, #80H

ACALL COMMAND

MOV A, #' '

ACALL DATA

MOV A, #81H

ACALL COMMAND

MOV A, #' '

ACALL DATA

ACALL DELAY

SJMP BLINK

COMMAND:

MOV P1, A

CLR P3.5

CLR P3.6

SETB P3.7

ACALL DELAY

CLR P3.7

RET

DATA:

MOV P1, A

SETB P3.5

CLR P3.6

SETB P3.7

ACALL DELAY

CLR P3.7

RET

DELAY:

MOV R6, #255

D1:

MOV R7, #255

D2:

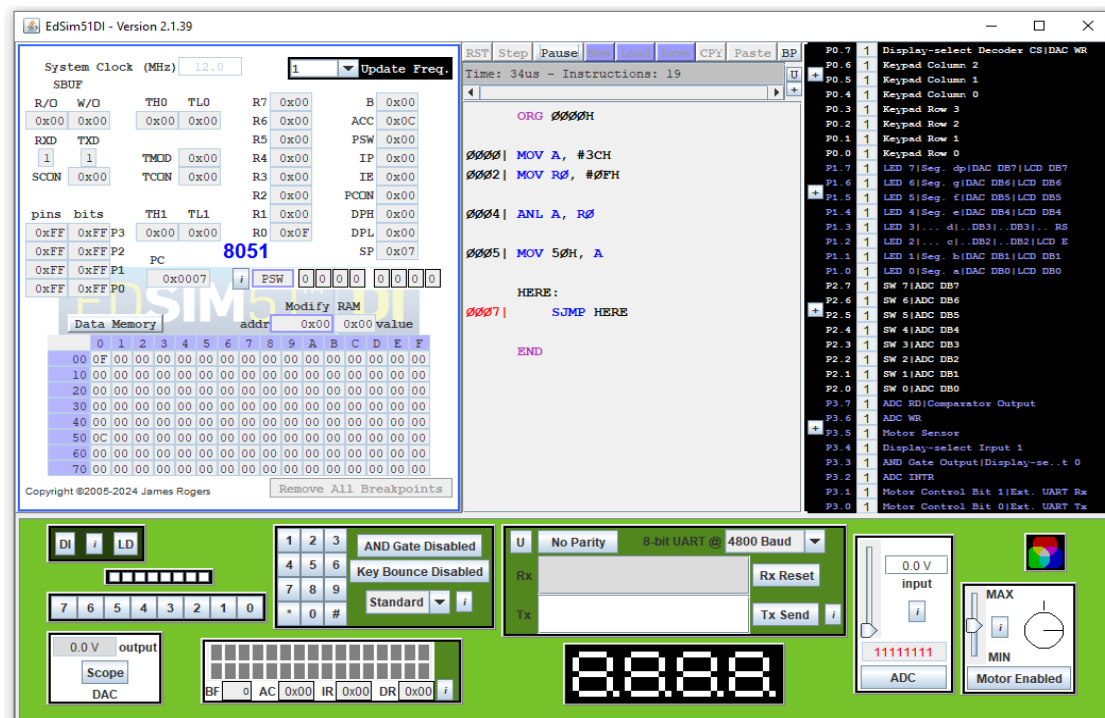
DJNZ R7, D2

DJNZ R6, D1

RET

END

Q. Write a program to perform AND operation



CODE:

ORG 0000H

MOV A, #3CH

MOV R0, #0FH

ANL A, R0

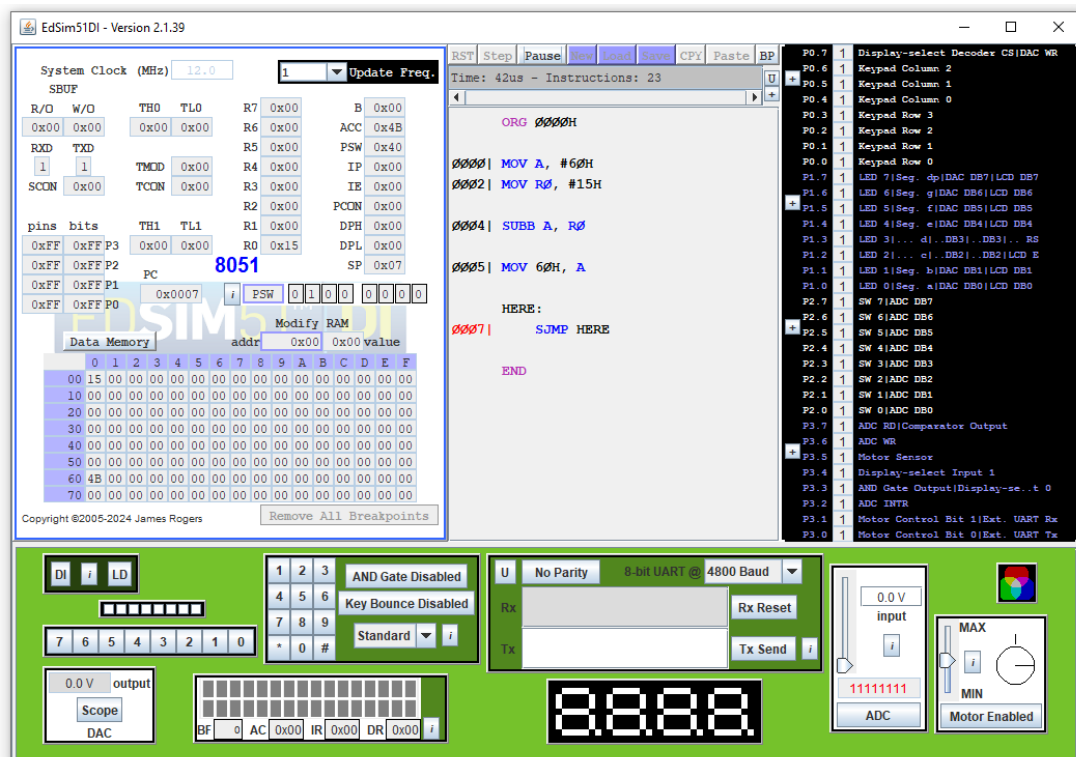
MOV 50H, A

HERE:

SJMP HERE

END

Q. Write a program to perform subtraction 20h FROM 50H



CODE:

```
ORG 0000H
```

```
MOV A, #60H
```

```
MOV R0, #15H
```

```
SUBB A, R0
```

```
MOV 60H, A
```

```
HERE:
```

```
    SJMP HERE
```

```
END
```